



OCTOBER 1, 2024

Town Hall Meeting: ChE Undergraduate Program

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UNIVERSITY OF
ILLINOIS CHICAGO

ABET Program Educational Objectives

PEO	Category	Description
1	Depth	In engineering practice, advanced study or original research, UIC Chemical Engineering bachelor's degree graduates will be effective in applying fundamental principles, scientific knowledge, rigorous analysis, creative design, and conceptual innovation.
2	Breadth	Based upon mastery of engineering in a broad, societal context, UIC Chemical Engineering graduates will have successful careers in the public or private sectors, or in the pursuit of graduate education.
3	Professionalism	In the service of industry, government, the engineering profession and society at large, graduates will function effectively in the complex modern work environment with clear communication, responsible teamwork, and high standards of ethics, professionalism, safety and protection of the environment.
4	Life-long learning	Based upon a rigorous undergraduate program that is innovative, challenging, open and supportive, graduates will enhance their skills and knowledge through life-long learning and demonstrate professional leadership.



ChE Curriculum:

Core Courses

Thermodynamics-Based Courses

- ★ ChE 201 – Introduction to thermodynamics
- ★ ChE 301 – Chemical Engineering Thermodynamics
- ChE 321 – Chemical Reaction Engineering

Transport-Based Courses

- ChE 311 – Transport Phenomena I (Fluid Mechanics)
- ChE 312 – Transport Phenomena II (Heat & Mass Transfer)
- ★ ChE 313 – Transport Phenomena III (Separations)

Process-Oriented Courses

- ChE 210 – Material & Energy Balances
- ChE 341 – Chemical Process Control
- ChE 396 – Senior Design I (Equipment Design)
- ChE 397 – Senior Design II (Design teams)

Unit Operations Laboratory

- ChE 381 – Chem Eng Lab I
- ChE 382 – Chem Eng Lab II

Numerical Analysis

- ChE 205- Numerical Methods in ChE

Molecular Engineering Courses

- ★ CHE 230 Molecular Engineering in CHE
- ★ CHE 330 Polymer Science

Engineering Education & Development

- ENGR100 -Engineering Orientation (Engineering Success Program, Guaranteed Paid Internship Program)
- ChE 499- Professional Development Seminar

ChE 400-level Elective
CHE 392:
Undergraduate
Research

Six Concentrations (15% of ChE students):

Biochemical Engineering	9 students
Process Automation	6
Energy & Environment	4
Nanotechnology	3
Polymers & Molecular Engineering	3
Entrepreneurship	0



Curriculum Improvement in progress...

Chemical
engineers are at
the forefront of
transforming our
future with
sustainable
solutions

PROGRAM REVISION IDEAS/ PROPOSALS:

(1)

ADD CHE 101 INTRODUCTION TO SUSTAINABLE
CHEMICAL ENGINEERING TO CORE CURRICULUM (+3
HOURS)

CREATE A NEW SELECTIVE OPTION: CHE 230
MOLECULAR SYSTEMS IN CHEMICAL ENGINEERING
OR CME 260 PROPERTIES OF MATERIALS (-3 HOURS)

(2)

REMOVE CHEMISTRY COURSE FROM MATH/SCIENCE
REQUIREMENT (-3 HOURS)

ADD SECOND TECHNICAL ELECTIVE TO CORE
REQUIREMENT (+3 HOURS)

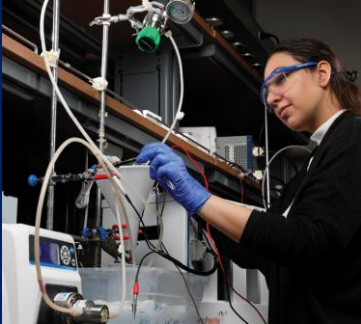
(3)

ADD DATA SCIENCE TO CHE CURRICULUM:

CREATE A NEW DATA SCIENCE CONCENTRATION



JOB OPPORTUNITIES



Chemical Engineering Employment Sectors

biotechnika
Your Bio Resource



Biotechnology



Chemical and allied
products



Process Plant
Construction



Pharmaceutical &
Toiletries



Water Industry



Food & Beverage
industries



Environment Protection
& Recycling



Energy (Oil,
Gas, & Nuclear)



Process plant &
Equipment
Manufacture



Materials (Plastics, dyes,
metals, paper, etc.)



Academics



IT



Consultancy



Others

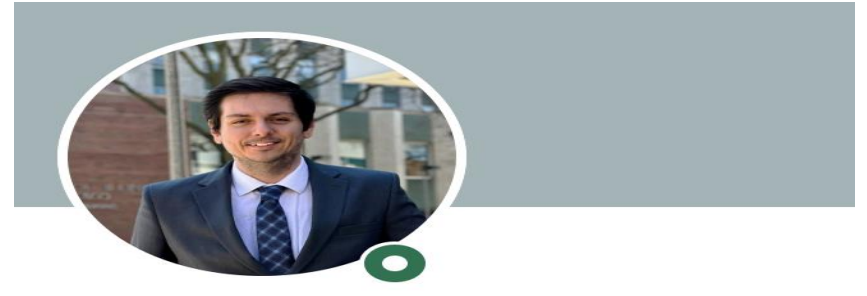


UIC Chemical Engineering graduates are in high demand



Ana Hernandez

Ana Hernandez (She/Her) · 1st
Process Manager @L'Oréal | People Leader |



Jordan Flanigan · 1st
Semiconductor Process Engineer at Intel



Aesha Talia · 1st
Product Manager at Medline Industries, Inc.



Emir Asani · 1st
Continuous Improvement Engineer at Mars Ice Cream



Courteney Bruce-Kotey · 1st
Process Engineer at Brindley Engineering Corporation



Paul Diaz · 1st
System Manager for digital I&C and mechanical equipment at Constellation Nuclear



Student - ChE Community Interaction

- WELCOME BACK PICNIC
- END-OF-YEAR BANQUET
- AICHE STUDENT CHAPTER

GENERAL MEETINGS, PROFESSIONAL & FUN ACTIVITIES
AICHE NATIONAL MEETING, NORTH-CENTRAL REGION
CHEM CAR, CHEM CUBE

- CHE EXPO
- UIC CHEMICAL ENGINEERING LINKEDIN GROUP

- PAST EVENTS:

CHE INDUSTRY DAY- PROFESSIONAL ENGINEER PANEL
DISCUSSION

CHE FORUM:

‘MINI-CONFERENCE’ WITH TECHNICAL SESSIONS
ALUMNI ‘MIXER’: STUDENTS & FACULTY INVITED